

Taeho Kim

[Google Scholar](#) | [Github](#) | [Personal Website](#)

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EDUCATION

University of Michigan–Ann Arbor

PhD candidate in Environmental Engineering

GPA: 4.0/4.0

Thesis: Decoding Plant–Hydrological Interactions across Climate-responsive Ecosystems: An Integrated Ecohydrological Framework Bridging Field Observations, Physical Processes, and Modeling

Advisor: Valeriy Ivanov

MI, United States

Aug. 2022 – Present

University of Seoul

Master of Science in Environmental Engineering

GPA: 4.5/4.5 (Top honor)

Thesis: Prediction of fecal pollution characteristics in streams using a hierarchical Bayesian network

Advisor: YoonKyung Cha

Seoul, South Korea

Mar. 2020 – Aug. 2021

University of Seoul

Bachelor of Science in Environmental Engineering

GPA: 4.27/4.5 (Top honor in the College of Urban Science – 1/309)

Thesis: Development of the water quality index of urban rivers considering land-use

Seoul, South Korea

Mar. 2014 – Feb. 2020

RESEARCH INTERESTS

Advancement of Ecohydrology for understanding water and carbon cycles.

Exploration of emerging patterns and biophysical processes in climate-sensitive regions.

Advancement of representations in land and Earth system models.

Application of advanced, scalable, and transferable ML/AI approaches.

RESEARCH EXPERIENCE

University of Michigan–Ann Arbor

Graduate Researcher, HYDROWIT

United States

Sep. 2022 – Present

- Analyze heterogeneous net radiation patterns in forest-tundra ecotone in the Ural mountainous area.
- Categorize diurnal sapflow patterns as Sapflow Functional Traits in the Amazon rainforest.
- Construct a predictive framework of sapflow measurement with causal interpretation.
- Identify and define extreme evapotranspiration as a novel environmental shock in ecosystems.
- Develop advanced ecohydrological models (ATS and tRIBS+VEGGIE) in the Amazon rainforest.

Oak Ridge National Lab

Graduate Research Intern, Watershed systems modeling group Aug. 2023; Jun.-July 2024; Jan. 2026– Present

United States

- Develop Lagrangian Parcel Tracking tool for Ecohydrology (EcoLPT).
- Variability in ecosystem resilience in response to hurricane disturbances.
- Develop surrogate AI models for complex landscapes with geometry-informed neural operators.

University of Seoul

Graduate Researcher, Water Environment Management Lab

South Korea

Jul. 2021 – Aug. 2022

- Constructed a module of deep Bayesian network predicting Chl-a from multispectral drone imagery.
- Constructed interpretable deep learning method for incomplete time-series data (RETAIN-D).
- Developed water quality index for source water and water purification plants.

Graduate Research Assistant, Water Environment Management Lab

Mar. 2020 – Jun. 2021

- Assessed fecal pollution characteristics in national streams with hierarchical Bayesian networks.
- Determined optimized locations for constructing new air quality monitoring sites in Seoul.
- Predicted eutrophication levels of unmonitored lakes using the transfer learning of remote sensing data.
- Applied interpretable machine learning methodology, SHAP (SHapley Additive exPlanations), to analyze EPT (Ephemeroptera, Plecoptera, and Trichoptera) species.
- Predicted total nitrogen loadings using the SPARROW (SPATIally Referenced Regressions On

Watershed attributes) model in Nakdong watershed.

Undergraduate Research Assistant, Water Environment Management Lab

Mar. 2018 – Feb. 2020

- Devised a water quality index methodology for urban streams using factor analysis.
- Developed dissolved oxygen prediction model for streams using the LSTM and Ensemble models.
- Analyzed the spatiotemporal patterns of TMDL water quality variables with load-duration curves.
- Identified relationships between social media text data and air quality through text analysis.

TEACHING AND MENTORING EXPERIENCE

Mentoring for Graduate Students

Academic Mentor

2024, 2026

Guided three Master students to successfully complete their research and submitted a journal article

- Research 2026: Global Daily 250 m Chlorophyll-a Dataset for Inland Lakes based on MODIS
- Thesis 2024: Interactions between sap flow dynamics in tree species and flooding in a floodplain forest: implications for biosphere–atmosphere processes in Santarém, Pará

Computational Methods for Engineers and Scientists (CEE 303: Prof. Rabab Haider)

Graduate Student Instructor

Jan. 2025 – Apr. 2025

Managed lab sessions and office hours.

Groundwater Hydrology (CEE 373: Prof. Brian Ellis)

Graduate Student Instructor

Sept. 2024 – Dec. 2024

Managed lab sessions (MODFLOW) and office hours.

Statistical Methods for Data Analysis and Uncertainty Modeling (CEE 373: Prof. Sabine Loos)

Graduate Student Instructor

Sept. 2023 – Dec. 2023

Managed office hours and supported class materials.

Engineering Statistics

Teaching Assistant

Mar. 2020 – Jun. 2020; Mar. 2021 – Jun. 2021

Answered course-related questions, created examples and homework programming codes.

Graduation Thesis Class for Undergraduate Students

Academic Mentor

2019, 2020, 2021

Guided seven undergraduates to successfully complete their bachelor's thesis and published a journal article with one of them.

- Thesis 2019: Analysis of trends in public interest for the fine dust (PM_{2.5} and PM₁₀)
- Thesis 2020: Pollution type classification of BOD and TP in national scale based on Load Duration Curve
- Thesis 2021: Prediction of TN load in Nakdong River using the SPARROW model

Computer Programming Study

Study Leader; Mentor

Aug. 2018 – Jun. 2020

Organized regular computer programming study groups participated by more than 20 members; taught Python to advanced ML/AI algorithms; advised on team projects and led two teams to win first place in the national data competitions.

HONORS & AWARDS

Graduate

Rackham Predoctoral Fellowship (\$90,000)	May. 2026 – May. 2027
Student-Identified Rackham Doctoral Intern Fellowship (\$30,000)	Jan. 2026 – Apr. 2026
Global and Regional Research and Internship Award (\$2,500)	July. 2025 – Aug. 2025
Rackham International Research Award (RIRA) (\$5,000)	July. 2024 – Aug. 2024
Rackham International Student Fellowship 2023-2024 (\$12,990)	Dec. 2023 – Aug. 2024
Graduate Student Research Grant (\$3,000/\$1,500)	July. 2024, July. 2023
Streeter Fellowship	Aug. 2022 – Apr. 2023
Outstanding Conference Poster Award	Nov. 2022
Certificate of Merit (Top honor) (GPA 4.5/4.5)	Aug. 2021
The Combined BS/MS Degree Scholarship (Full tuition)	Mar. 2020 – Aug. 2021
Instruction Assistance Scholarship	Mar. 2021 – Aug. 2021

Undergraduate

Certificate of Merit (Top honor – 1/309) in the College of Urban science (GPA 4.27/4.5)	Feb. 2020
Scholarship for Excellent Achievement (Full tuition)	Mar. 2014 – Feb. 2020
University Overseas Support Project Scholarship (Nepal)	Aug. 2019
Academic Achievement Award (GPA 4.5/4.5)	Spring 2018, Fall 2018, & Spring 2019
Scholarship for Tutoring (Calculus)	Sept. 2015 – Dec. 2015

Competitions

<i>Grand Prize</i> , National Big Data Contest (Dacon)	Aug. 2020
(Topic: Prediction of credit card usage amount in Jeju, using AI algorithms)	
<i>Grand Prize</i> , National Environment Big Data Contest (K-water)	Oct. 2019
(Project: Predicting alien fish species density using deep learning algorithm in Han River, South Korea.)	

PUBLICATIONS

- [1] **Kim, T.**, Le, P., Coon, E., Painter, S., Tran, V., Taylor, T., & Ivanov, V.. EcoLPT v1.0: Lagrangian Parcel Tracking tool for Ecohydrology, *Geoscientific Model Development (In preparation)*.
- [2] **Kim, T.**, Taylor, T., Agee, E., Restrepo-Coupe, N., Campos, K., Prohaska, N., Albert, L., Figueira, M., Moura, J., Ferreira, S., Senna, R., Oliveira Jr., R., Ferreira, M., Rodrigues, J., Souza, T., & Ivanov, V. Dynamic interpretation of tree-level sapflow response to environmental changes in the Amazon Rainforest. *Proceedings of the National Academy of Sciences (In review)*.
- [3] Yang, S., Lee, H., Lee, G., Gu, T., Shin, J., **Kim, T.**[†], & Cha, Y. (2026) GLCD250-MOD: Global Daily 250 m Chlorophyll-a Dataset for Inland Lakes based on MODIS, *Earth System Science Data (In review)*.
- [4] Nelson-Mercer, B., Bricker, J., Notaro, M., Bechle, A., Hong, Y., Shriberg, M., Berkel, D., Li, M., Mullenbach, L., **Kim, T.**, & Manome, A. (2026) Development and validation of a Lake Michigan compound flood model, *Journal of Flood Risk Management (In review)*.
- [5] Tran, V., Xu, D., Lu, D., Le, P., Wang, J., Leung, L., Coupe, N., Taylor, T., Zhou, W., **Kim, T.**, Fatichi, S., & Ivanov, V. (2026) Dynamic Controls on Global Terrestrial Evapotranspiration Revealed by a Differentiable Physics–Machine Learning Framework, *Journal of Geophysical Research: Machine Learning and Computation (In review)*.
- [6] Ivanov, V.¹, **Kim, T.**¹, Taylor, T., Agee, E., Albert, L. P., Alves, L. F., Assis, P. I. L. S., Campos, K. S., Restrepo-Coupe, N., Ewing, G., Figueira, M., Franco, T. S. B. S., Huang, W., Oliveira Júnior, R. C., Machado, W., Moura, J. M. S., Prohaska, N., Oliveira, R. S., Rodrigues, J. V. F. C., Saleska, S. R., Santos, D. B. dos, Tapajos, R., & Zhang, L. Identification of Tree Water Uptake Strategies in the Amazon Rainforest as Sapflow Functional Types, *New Phytologist (In review)*.
- [7] **Kim, T.**, Zhou, W., Tran, V., Wang, F., Zhu, M., Sheshukov, A., Zhang, T., Liu, D., Mazepa, V., Sokolov, A., Valdayskikh, V., & Ivanov, V. (2025). Biases in radiative flux observations due to precipitation across the Arctic forest-tundra ecotone, *Agricultural and Forest Meteorology*, 374, 110814.
- [8] **Kim, T.**, Lee, H., Yang, S., Lee, G., Shin, J., & Cha, Y. (2025). Long-term Spatiotemporal Variability and Regime Classification of Chlorophyll-a Concentrations in Lake Erie Using Satellite Products. *Harmful Algae*, 102896.
- [9] Tran V., **Kim, T.**, Xu, D., Nguyen T., & Ivanov, V. (2025). CAMELSH: A Large-Sample Hourly Hydrometeorological Dataset and Attributes at Watershed-Scale for CONUS, *Scientific Data*, 12(1), 1307.
- [10] Tran V., **Kim, T.**, Xu, D., Tran, H., Le, M., Policelli, F., Tran, T., Kim, J., Wright, D., Restrepo, P., & Ivanov, V. (2025). AI improves the accuracy, reliability, and economic value of continental-scale flood predictions, *AGU Advances*, e2025AV001678.
- [11] Lee, G., Shin, J., Kim, Y., **Kim, T.**, Han, E., Yu, C., & Cha, Y. (2025) Deep learning-based prediction of exceeding the criteria for river chlorophyll-a concentrations using high-frequency data from a sensor network, *Environmental Engineering Research*, 302.
- [12] Nelson-Mercer, B., **Kim, T.**, Tran, V., & Ivanov, V. (2025). Pluvial flood impacts and policyholder responses throughout the United States, *NPJ Natural Hazard* 2, 8.
- [13] Han, J., **Kim, T.**, Lee, S., Kang, T., & Im, J. (2024). Machine learning and explainable AI for Chlorophyll-a prediction in Namhan River watershed, South Korea, *Ecological Indicators*, 166, 112361.

- [14] Tran V., Zhou W., **Kim T.**, Mazepa, V., Valdayskikh, V., & Ivanov, V. (2024). Daily station-level records of air temperature, snow depth, and ground temperature in the Northern Hemisphere, *Scientific Data*, 11(1), 645.
- [15] Shin, J., Lee, G., **Kim, T.**, Cho, K., Hong, S., Kwon, D., Pyo, J., & Cha, Y. (2023). Deep learning-based efficient drone-borne sensing of cyanobacterial blooms using a clique-based feature extraction approach. *Science of Total Environment*, 169540.
- [16] Lee, D., Shin, J., **Kim, T.**, Lee, S., Kim, D., Park, Y., Cha, T., & Cha, Y. (2023). Hybrid model for daily streamflow and phosphorus load prediction. *Water Science and Technology*, 88(4), 975-990.
- [17] **Kim, T.**, Shin, J., & Cha, Y. (2023). Incorporation of feature engineering and attention mechanisms into deep learning models to develop an early warning system for harmful algal blooms. *Journal of Cleaner Production*, 137564.
- [18] **Kim, T.**, Shin, J., Kim, Y., Lee, D., Beak, J., Lee, D., & Cha, Y. (2023). Expansion of roadside air pollution monitoring network in Seoul using learning-based optimization method. *Journal of Korean Society of Environmental Engineers*, 45(3), 148-160.
- [19] Kang C., Shin, J., Cha, Y., Kim, M., Choi, M., **Kim, T.**, Park, Y., & Choi, Y. (2023). Machine learning-guided prediction of potential engineering targets for microbial production of lycopene. *Bioresource Technology*, 128455 [Cover].
- [20] Lee, J., Lee, J., Cha, Y., Cho, S., **Kim, T.**, Cha, Y., & Koo, J. (2022). Development of customized water quality index for medium-and long-term evaluation of urban river water quality. *Water Supply*, 22(7), 6338-6355.
- [21] **Kim, T.**, Shin, J., Lee, D., Kim, Y., Na, E., Park, J., Lim, C., & Cha, Y. (2022). Simultaneous feature engineering and interpretation for harmful algal bloom forecasts with a deep learning approach. *Water Research*, 118289.
- [22] Kim, Y., **Kim, T.**, Shin, J., Lee, D., Park, Y., Kim, Y., Cha, Y. (2022). Validity evaluation of a machine learning model for chlorophyll-a retrieval from Sentinel-2 (MSI) in inland and coastal waters. *Ecological Indicators*, 108737.
- [23] **Kim, T.**, Lee, D., Shin, J., Kim, Y., & Cha, Y. (2022). Learning hierarchical Bayesian networks to assess the interaction effects of controlling factors on spatiotemporal patterns of fecal pollution in streams. *Science of Total Environment*, 152520.
- [24] Cha, Y., Shin, J., Go, B., Lee, D. S., Kim, Y., **Kim, T.**, & Park, Y. S. (2021). An interpretable machine learning method for supporting ecosystem management: Application to species distribution models of freshwater macroinvertebrates. *Journal of Environmental Management*, 291, 112719.
- [25] Kim, Y., **Kim, T.**, Shin, J., Go, B., Lee, M., Lee, J., Cho, K., & Cha, Y. (2021). Forecasting abrupt depletion of dissolved oxygen in urban streams using discontinuously measured hourly time-series data. *Water Resources Research*, 57(4), e2020WR029188.
- [26] Shin, J., Yoon, S., Kim, Y., **Kim, T.**, Go, B., & Cha, Y. (2021). Effects of class imbalance on resampling and ensemble learning for improved prediction of cyanobacteria blooms. *Ecological Informatics*, 61, 101202.
- [27] **Kim, T.**, Kim, Y., Shin, J., Go, B., & Cha, Y. (2020). Assessing land-cover effects on stream water quality in metropolitan areas using the water quality index. *Water*, 12(11), 3294. (Editor's choice)
- [28] Lee, J., Ha, H., Lee, M., **Kim, T.**, Cha, Y., & Koo, J. (2020). Assessment of water quality of major tributaries in Seoul using water quality index and cluster analysis. *Journal of Korean Society of Environmental Engineers*, 42(10), 452-462.

CONFERENCES

- [1] **Kim T.**, Coon E., Painter S., Tran V., Taylor T., Ivanov V., & Le P. EcoLPT v1.0: Lagrangian Parcel Tracking tool for Ecohydrology, American Geophysical Union Winter Conference, 2026.
- [2] **Kim T.**, Yang H., Lee J., & Ivanov V. HT-PTF: A Hydro-Topographical Pedotransfer Function Based on Operator Learning for Continuous Soil Water Retention Prediction, American Geophysical Union Winter Conference, 2026.
- [3] Yang H., Li Z., Lee J., **Kim T.**, Tran V., Lu D., Dias P., Painter S., & Le P. Geometry-Aware Neural Operators for Transferable Storm Surge Prediction, American Geophysical Union Winter Conference, 2026.
- [4] Li Z., Yang H., Dias P., Faxon A., McCarthy M., **Kim T.**, Sulman B., Painter S., Tran V., Lu D., & Le P. Divergent Controls on Coastal Wetland Resistance and Recovery to Tropical Cyclones, American

Geophysical Union Winter Conference, 2026.

- [5] **Kim T.**, Taylor T., Agee E., Restrepo-Coupe N., Campos K., Prohaska N., Albert L., Figueira M., Moura J., Ferreira S., Senna R., Oliveira Jr. R., Ferreira M., Rodrigues J., Souza T., & Ivanov V. Dynamic interpretation of tree-level sapflow response to environmental changes in the Amazon Rainforest, American Geophysical Union Winter Conference, 2025.
- [6] **Kim, T.**, Agee, E., Huang, W., Taylor, T., & Ivanov, V. Identifying tree water uptake strategies in the Amazon rainforest through sapflow functional traits, American Geophysical Union Winter Conference, 2024.
- [7] Ivanov, V., Taylor, T., Albert, L., Saleska, S., Braun, E., Southwick, C., Garcia, M., Souza T., Coupe, N.R. ... **Kim, T.**, Santos, A. Quantifying tree hydraulic function across distinct soil hydrologic environments with measurements and modeling in the Amazon, American Geophysical Union Winter Conference, 2023.
- [8] **Kim, T.**, Zhou, W., Ivanov, V., Shivatov, S., Wang, J., Sheshukov, A., Liu, D., & Mazepa, V.. Variations of Net Radiation in High-Latitude Forest and Tundra. American Geophysical Union Winter Conference, 2023.
- [9] **Kim, T.**, Lee, D., Shin, J., Kim, Y., & Cha, Y. Development of deep Bayesian networks for environment modeling. Korean Society of Environmental Engineers Winter Conference, 2021.
- [10] Lee, D., **Kim, T.**, Shin, J., Kim, Y., & Cha, Y. Assessing real-time feature contribution change of environment variables in the occurrence of algal bloom warning. Korean Society of Environmental Engineers Winter Conference, 2021 [*Award 9 Nov. 2022*].
- [11] **Kim, T.**, Shin, J., Kim, Y., Go, B., Beak, J., Lee, D., Lee., D., & Cha, Y. Establishment of optimization methodology for expanding roadside air quality monitoring network (AQMN) in Seoul. Korean Society of Environmental Engineers Winter Conference, 2020.
- [12] Go, B., Shin., J., Kim, Y., **Kim, T.**, & Cha, Y. The effect of hyperparameter optimization method and generalization performance estimation method of machine learning model on the performance of river water quality forecasting model, Korean Society of Environmental Engineers Winter Conference, 2020.
- [13] Shin, J., Kim, Y., **Kim, T.**, Go, B., Cha, Y., & Yoon, S. Improving the predicting performance of cyanobacteria bloom occurrence model using data resampling and ensemble learning. Korean Society of Environmental Engineers Winter Conference, 2020.
- [14] Kim, Y., Shin, J., **Kim, T.**, Go, B., & Cha, Y. A study on the interpretable machine learning techniques of river dissolved oxygen forecasting model based on machine learning. Korean Society of Environmental Engineers Winter Conference, 2020.
- [15] Shin, J., Cha, Y., **Kim, T.**, Lee, M., Kil, H., Ha, H., & Lee, J. Characterization of urban stream water quality using water quality index, Geophysical Research Abstracts, 2019.
- [16] **Kim, T.**, Im, I., Youn, S., Jeon, H., Kim, Y., Shin, J., & Cha, Y. Identification of major water quality clusters using integrated water quality index in urban rivers. Korean Society of Environmental Engineers Winter Conference, 2018.

PROJECTS

- [1] Advancing Coastal Resilience with Multi-modal Generative Artificial Intelligence, Department of Energy, Graduate Researcher, Jan 2026 –Present
- [2] A perfect storm: will the double-impact of 2023/24 El Nino drought and forest degradation induce a local tipping-point onset in the eastern Amazon?, National Science Foundation, Graduate Researcher, Feb. 2024 – 2026.
- [3] Interactions of natural and social systems with climate change, globalization, and infrastructure development in the Arctic, National Science Foundation (NSF), Graduate Researcher, Dec. 2023 – 2026.
- [4] Cascade “Ecohydromics” in the Amazonian Headwater System, National Science Foundation (NSF), Graduate Researcher, Sept. 2022 – 2026.
- [5] Development of ecological prediction models and health assessment methods in lentic ecosystems, Korea Environmental Industry and Technology Institute (KEITI), Graduate Researcher, May 2020 – Aug. 2022.
- [6] Development of the big data platform and standardization techniques for the management of waterworks and sewer systems, Environmental Industry and Technology Institute (KEITI), Graduate Researcher, Apr. 2020 – Aug. 2022.
- [7] Development of a modularized AI-based model for the multiscale analysis of multimodal environmental data, Ministry of Science and ICT (MSIT), Graduate Researcher, Mar. 2020 – Aug. 2022.

- [8] Developing optimization methodology of roadside air quality monitoring network (AQMN) for supporting fine dust management strategy in Seoul, Seoul Green Environmental Center (SGEC), Graduate Research Assistant, Mar. 2020 – Jan. 2021.
- [9] Development of prediction model for next generation water quality and aquatic ecosystem (II) – Assessment of aquatic ecological models using water quality and aquatic ecosystem monitoring data, Undergraduate Research Assistant, Oct. 2018 – Oct. 2019.
- [10] Development of core element technology for Seoul-type river water quality management system based on big data to respond to climate change, Undergraduate Research Assistant, Oct. 2018 – Aug. 2019.

PATENTS

- [1] Lee, J., Lee, M., Lee, M., Lee, J., Kim, H., Cha, Y., & **Kim, T.** (2025) Extraction of major water quality management variables for rivers using factor analysis and the development of customized water quality index reflecting regional differences, Korea Patent, 10-2841247.

RESEARCH & FIELD TRIPS

Oak Ridge National Lab (ORNL), Tennessee Aug. 2023; Jun.-July 2024; Jan.-Apr 2026
 Develop Lagrangian Parcel Tracking tool for Ecohydrology (EcoLPT).
 Learn Advanced Terrestrial Simulator (ATS) from Watershed Systems Modeling Group advised by Dr. Ethan Coon, Dr. Scott Painter, and Dr. Phong Le.

Santarém, K67, Amazon in Brazil Aug. 2023; May & Aug. 2024; Aug. 2025
 Adventure hillslope system (Plateau and Baixio) and setup field monitoring systems on soil, trees, and atmosphere.

SKILLS

Software

Python, R, MATLAB, C/C++, ArcGIS, QGIS, Google Earth Engine

Modeling Techniques

Advanced Terrestrial Simulator (ATS), tRIBS+VEGGIE, SWAT, SWIM, OpenFOAM, EcoLPT, Bayesian statistics, Bayesian network, Factor analysis, Structural equation modeling, Deep learning, Machine learning, Genetic algorithm.

National Technical Qualification Certificates

Big-Data Analysis Engineer (08/2021), Engineer Water Pollution Environmental (06/2020), & Craftsman Environmental (06/2017).

Languages

Korean (native), English (advanced), French (elementary), & Mandarin (elementary).

ACTIVITIES & INTERESTS

Cantabile, University of Seoul music club

Club Member

Mar. 2020 – Jul. 2021

Performed several orchestras and ensembles with playing cello.

ECO Road, University association environment club

Club President

Mar. 2015 – May 2016

Arranged club plans and hold environmental seminars and performances.

Other Interests

Cello, Tennis, Taekwondo, and Trekking

MILITARY SERVICE

Republic of Korea Army

May 2016 – Feb. 2018

Completed military service as a unit supply sergeant.

REFERENCES

Valeriy Ivanov, Professor
Civil and Environmental Engineering,
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